Activities
1) Find 10 different model providers and their models released in last 12 months

- **OpenAI**
  - **Model:** GPT-6 Astra
  - **Release Date:** 3 Sep 2026
  - **Cut off Date:** 30 Apr 2026
  - **Cost of tokens:** $10 input / $50 output per 1M tokens
  - **Context Window:** 1.05M tokens (128K output)
  - **Type:** Closed, reasoning, text and image input
  - **When to use this model:** Hardest reasoning, coding, computer-use and research tasks
- **Anthropic**
  - **Model:** Claude Opus 5.5
  - **Release Date:** 22 Sep 2026
  - **Cut off Date:** Jun 2026
  - **Cost of tokens:** $4 input / $20 output per 1M tokens
  - **Context Window:** 1M tokens (128K output)
  - **Type:** Closed, adaptive thinking, text and image input
  - **When to use this model:** Long-running agentic coding, knowledge work and computer use
- **Google**
  - **Model:** Gemini 4 Argon
  - **Release Date:** 13 Oct 2026
  - **Cut off Date:** Mar 2026
  - **Cost of tokens:** $2 input / $10 output per 1M tokens until 30 Nov 2026, then $4 / $20
  - **Context Window:** 1M tokens (65K output)
  - **Type:** Closed, MoE, reasoning, text, image, video, audio and PDF input
  - **When to use this model:** Complex software engineering, math and science, long multi-step agents
- **xAI**
  - **Model:** Grok 4.7
  - **Release Date:** Sep 2026
  - **Cut off Date:** May 2026
  - **Cost of tokens:** $2 input / $6 output per 1M tokens (under 200K prompt); $4 / $12 above
  - **Context Window:** 500K tokens
  - **Type:** Closed, reasoning
  - **When to use this model:** Coding and agentic work where low output cost matters

- **Meta**
  - **Model:** Muse Wonder (preview)
  - **Release Date:** 12 Oct 2026
  - **Cut off Date:** Not published
  - **Cost of tokens:** $4 input / $20 output per 1M tokens
  - **Context Window:** 1M tokens
  - **Type:** Closed, multimodal, reasoning
  - **When to use this model:** Meta's top model, when quality matters more than cost
- **Mistral AI**
  - **Model:** Mistral Large 4
  - **Release Date:** 6 Oct 2026
  - **Cut off Date:** Not published
  - **Cost of tokens:** $0.70 input / $2.10 output per 1M tokens
  - **Context Window:** 1M tokens
  - **Type:** Open-weight, MoE (~1T total / 52B active), multimodal, reasoning
  - **When to use this model:** Frontier-class multimodal reasoning at low cost, or a large model to self-host
- **DeepSeek**
  - **Model:** DeepSeek-V4-Pro
  - **Release Date:** 13 Aug 2026
  - **Cut off Date:** Not published
  - **Cost of tokens:** $1.32 input / $3.96 output per 1M tokens at peak; half price off-peak
  - **Context Window:** 1M tokens (384K output)
  - **Type:** Open-weight (MIT), MoE (1.6T / 49B active), text only
  - **When to use this model:** Strong agents, coding and reasoning at a low price
- **Alibaba (Qwen)**
  - **Model:** Qwen3.8-Max
  - **Release Date:** 3 Aug 2026
  - **Cut off Date:** Not published
  - **Cost of tokens:** $2 input / $6 output per 1M tokens
  - **Context Window:** 1M tokens (128K output)
  - **Type:** Hosted, MoE (2.4T / 95B active), multimodal
  - **When to use this model:** Hardest coding, long-horizon agent and multimodal work
- **Moonshot AI (Kimi)**
  - **Model:** Kimi K3.1 Pro
  - **Release Date:** 23 Oct 2026
  - **Cut off Date:** Not published

    - **Cost of tokens:** $2 input / $10 output per 1M tokens
    - **Context Window:** 1M tokens
    - **Type:** Open-weight, MoE (2.8T / 104B active), text, image and video input
    - **When to use this model:** Long-horizon coding and agentic knowledge work
- **Zhipu AI (Z.ai)**
    - **Model:** GLM-5.3
    - **Release Date:** 18 Aug 2026
    - **Cut off Date:** Not published
    - **Cost of tokens:** $1.40 input / $4.40 output per 1M tokens
    - **Context Window:** 1M tokens (128K output)
    - **Type:** Open-weight, MoE (~753B), text only, reasoning
    - **When to use this model:** Agentic coding at a lower price than the US frontier models

2) Read 2–3 articles about how transformer works?
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Read the articles-**Articles**

- **The Illustrated Transformer** by Jay Alammar: https://jalammar.github.io/illustrated-transformer/
    - The best-known visual walkthrough, good for self-attention step by step.
- **Generative AI exists because of the transformer** by the Financial Times: https://ig.ft.com/generative-ai/
    - A short, animated explainer for non-engineers, good for embeddings and attention.

      —-----The examples are great to understand in the above article—--
- **Transformers, the tech behind LLMs** by 3Blue1Brown (video and article): https://www.3blue1brown.com/lessons/gpt